

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

yoot@my.erau.edu

| [Website](#)

| [Linkedin](#)

| [Google Scholar](#)

Education

Embry-Riddle Aeronautical University
M.S. in Aviation
Specialization in Aviation Safety Systems

Daytona Beach, FL
Aug 2025–Present
GPA 4.0/4.0

Embry-Riddle Aeronautical University
B.S. in Aeronautical Science
Minor in Air Transportation

Daytona Beach, FL
May 2023
GPA 3.85/4.0

Research Interests

Flight Safety Analytics; ADS-B Analysis; Approach Stability Assessment; Terminal Airspace Operations

Research Experience

Graduate Researcher

School of Graduate Studies, College of Aviation

Advisor: Dr. João Souza Dias Garcia

Sep 2025–Present

- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases

Graduate Researcher

Applied Aerospace Research & Development Lab

Advisor: Dr. Ryan Wallace

Feb 2026–Present

- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[1] Yoo, T. & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. (Accepted). American Institute of Aeronautics and Astronautics AVIATION Forum.

Presentations

- Characterizing Air Traffic Density Using Nationwide ADS-B Data. (Poster). Embry-Riddle Aeronautical University Discovery Day, 2026.
- Data-Driven Framework for Assessing Wind Impacts on Approach Stability in Flight Training. (Poster). BCAAS Third Annual Safety Research Symposium, 2026
- Approach Stability Assessment in Flight Training Using ADS-B Data. (Poster). Embry-Riddle Aeronautical University Student Research Symposium, 2025.

Teaching Experience

Graduate Teaching Assistant

Supervisor: Dr. Dahai Liu

Embry-Riddle Aeronautical University

Dec 2025–Present

Teaching support for:

- Statistical Analysis for Aviation/Aerospace
- Research Methods in Aviation/Aerospace
- Graduate Capstone Research Project

Flight Instructor

Instructional areas include:

- Private Pilot Certificate
- Instrument Rating
- Commercial Pilot Certificate
- Certified Flight Instructor
- Certified Flight Instructor-Instrument
- Flight Review and Instrument Proficiency Check
- Complex Endorsement

Aviators Academy
Nov 2023–July 2025

Honors and Awards

Best Poster Award, Embry-Riddle Aeronautical University	April 2026
1st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University	November 2025
Magna Cum Laude, Embry-Riddle Aeronautical University	May 2023
Deans List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University	

Certifications and Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane
Flight Instructor – ASEL, AMEL, Instrument Airplane
Ground Instructor – Advanced, Instrument
Remote Pilot – sUAS

Programming Skills

Python (pandas, traffic, scikit-learn, Matplotlib)
SPSS (PROCESS macro)

Professional Memberships

American Institute of Aeronautics and Astronautics
University Aviation Association
Royal Aeronautical Society

Leadership & Service

Secretary Korean Student Association	Embry-Riddle Aeronautical University 2019–2020
Administrative Specialist; Squad Commander Republic of Korea Military Service	Korea Defense Communication Command 2017–2019
Vice Captain Table Tennis Club	University of North Dakota 2016–2017